CSCE 222: Discrete Structures - Final Exam Reference Guide (Rosen 9th Ed.)

LOGIC & EQUIVALENCES

Operators: $\neg$ (not), $\wedge$ (and), $\vee$ (or), $\oplus$ (xor), $\rightarrow$ (if), $\leftrightarrow$ (iff).
Order: $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$ (High to Low).
Equivalences:

- $p \rightarrow q \equiv \neg p \vee q$
- $\neg(p \wedge q) \equiv \neg p \vee \neg q$ (De Morgan)
- $p \leftrightarrow q \equiv (\neg p \vee q) \wedge (\neg q \vee p)$
- **Identity:** $p \wedge T \equiv p$; $p \vee F \equiv p$
- **Domination:** $p \vee T \equiv T$; $p \wedge F \equiv F$
- **Absorption:** $p \vee (p \wedge q) \equiv p$; $p \wedge (p \vee q) \equiv p$

Fallacies:

- **Affirming Consequent:** $(p \rightarrow q) \wedge q \therefore p$ (Error)
- **Denying Antecedent:** $(p \rightarrow q) \wedge \neg p \therefore \neg q$ (Error)

RULES OF INFERENCE

$p, p \rightarrow q \therefore q$	Modus Ponens
$\neg q, p \rightarrow q \therefore \neg p$	Modus Tollens
$p \rightarrow q, q \rightarrow r \therefore p \rightarrow r$	Hypo. Syllogism
$p \vee q, \neg p \therefore q$	Disj. Syllogism
$p \therefore p \vee q$	Addition
$p \wedge q \therefore p$	Simplification
$p, q \therefore p \wedge q$	Conjunction
$p \vee q, \neg p \vee r \therefore q \vee r$	Resolution

PREDICATE LOGIC

Universal: $\forall x P(x)$ (True for all x in domain)
Existential: $\exists x P(x)$ (True for at least one x)
Negation:
$\neg \forall x P(x) \equiv \exists x \neg P(x)$
$\neg \exists x P(x) \equiv \forall x \neg P(x)$

PROOF TECHNIQUES

Direct: Assume p, show q.
Contrapositive: Assume $\neg q$, show $\neg p$.
Contradiction: Assume $p \wedge \neg q$, derive F.
Cases: Prove $(p_1 \vee p_2 \vee \cdots \vee p_n) \rightarrow q$ by proving $p_i \rightarrow q$ for each i.

SETS & CARDINALITY

$A \subseteq B \iff \forall x(x \in A \rightarrow x \in B)$
Operations: $A \cup B$ (or), $A \cap B$ (and), $A - B$ (A not in B).
Complement: $\overline{A} = U - A$.
Power Set: $P(A) = \{S \mid S \subseteq A\}$, $|P(A)| = 2^{|A|}$.
Cartesian: $A \times B = \{(a, b) \mid a \in A, b \in B\}$, $|A \times B| = |A||B|$.
Inclusion-Exclusion: $|A \cup B| = |A| + |B| - |A \cap B|$.

FUNCTIONS

Function: $f : A \rightarrow B$. A is domain, B is codomain.
Injective (1-1): $f(a) = f(b) \Rightarrow a = b$.
Surjective (Onto): $\forall y \in B, \exists x \in A, f(x) = y$.
Bijective: Both injective and surjective.
Inverse: f^{-1} exists iff f is a bijection.
Composition: $(f \circ g)(x) = f(g(x))$.

NUMBER THEORY

Divisibility: $a \mid b \iff b = ak$ for some integer k.
Division Alg: $a = dq + r$, $0 \leq r < d$.
Primes: $n > 1$, only factors are $1, n$.
Fund. Thm. Arithmetic: $n > 1$ has unique prime factorization.
GCD/LCM: $\gcd(a, b) \cdot \mathrm{lcm}(a, b) = ab$.

ALGORITHMS & COMPLEXITY

Big-O: $f(x) \leq Cg(x)$ for $x > k$.
Growth: $1 < \log n < n < n \log n < n^2 < 2^n < n! < n^n$.
Linear Search: $O(n)$. **Binary Search:** $O(\log n)$.
Bubble Sort: $\Theta(n^2)$. **Insertion Sort:** $O(n^2)$.

SEQUENCES & SUMS

Arithmetic: $a_n = a + (n-1)d$. **Sum:** $S_n = \frac{n(2a+(n-1)d)}{2}$.
Geometric: $a_n = ar^{n-1}$. **Sum:** $S_n = \frac{a(r^n-1)}{r-1}$ $(r \neq 1)$.
Series: $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$, $\sum i^2 = \frac{n(n+1)(2n+1)}{6}$.

COUNTING

Product Rule: $n_1 \cdot n_2$ ways for independent tasks.
Sum Rule: $n_1 + n_2$ ways for disjoint tasks.
Pigeonhole: N objects, k boxes $\Rightarrow$ at least one box has $\lceil N/k \rceil$.
 # Functions $A \rightarrow B$: $|B|^{|A|}$.
 # Injective $A \rightarrow B$: $P(|B|, |A|)$.

PERMS & COMBS

Permutations: $P(n, r) = \frac{n!}{(n-r)!}$.
Combinations: $C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$.
Binomial Thm: $(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$.
Pascal's Identity: $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$.
Vandermonde's: $\binom{m+n}{r} = \sum \binom{m}{r-k}\binom{n}{k}$.

RELATIONS

$R \subseteq A \times A$:

- **Reflexive:** $\forall a, (a, a) \in R$
- **Symmetric:** $(a, b) \in R \Rightarrow (b, a) \in R$
- **Antisymmetric:** $(a, b), (b, a) \in R \Rightarrow a = b$
- **Transitive:** $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$

Equiv. Relation: Refl, Symm, Trans.
Partial Order (Poset): Refl, Antisymm, Trans.
 # Relations on n elements: 2^{n^2}. **Reflexive:** $2^{n(n-1)}$.

RECURRENCE RELATIONS

Lin. Homog. order k: $a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k}$.
Char. Eq: $r^k - c_1 r^{k-1} - \cdots - c_k = 0$.
1. **Distinct roots** r_1, r_2: $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$.
2. **Repeated root** r_1: $a_n = (\alpha_1 + \alpha_2 n) r_1^n$.
Solving: 1. Find char eq. 2. Find roots. 3. Use IC to solve α_i.

INDUCTION TEMPLATE (EXACT)

Goal: $\forall n \geq a, P(n)$
Basis: Show $P(a)$ holds.
Inductive: 1. Assume $P(k)$ holds for $k \geq a$ (**I.H.**).
2. Show $P(k + 1)$ holds.
3. Use $P(k)$ to transform LH expression $\rightarrow$ RH.
Conclusion: By induction, $P(n)$ is true $\forall n \geq a$.

STRONG INDUCTION

Basis: Prove $P(a), \ldots, P(a + j)$ as needed.
Assume: $P(a) \wedge P(a + 1) \wedge \cdots \wedge P(k)$ holds.
Show: $P(k + 1)$ holds using any/all previous values.

QUICK STRATEGY

Equality: LHS $\rightarrow$ RHS using IH.
Inequality: Use IH + monotonic bounds $(k+1 > k)$.
Divisibility: Rewrite as $d \cdot (\ldots)$ using IH multiple.
Counting: Product (sequential) vs Sum (choice).
Binomial: $\binom{n}{k} = \binom{n}{n-k}$.